Harmonic Optimal Power Flow with Transformer Excitation

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***Abstract*—Harmonics pose significant challenges to the operation of power systems, and active filters are a key technology to mitigate impacts. When there are multiple active filters in a network, they can be coordinated to achieve specific goals, e.g., provide maximal compensation at the point of common coupling. Mathematical optimization of power systems with representation of harmonics, i.e., harmonic optimal power flow, is a framework that develops solutions to such decision problems. This paper presents a formulation for harmonic optimal power flow in the rectangular current-voltage variable space, and solves it as a continuous nonlinear optimization problem. A key novelty of this work is the detailed treatment of transformers, and transformer harmonics due to saturation. Normally, transformer excitation is modeled in the time domain, so the inclusion of these effects in frequency-domain optimization models poses a nontrivial challenge, which is addressed here through a conversion to empirical multivariate nonlinear continuous functions (splines). The model is demonstrated numerically on a real-world test case.**

***Index Terms*—Harmonic optimal power flow, transformer excitation, nonlinear programming, mathematical optimization.**

I. Introduction

A. Background and Motivation

Harmonics in power systems result in a deviation from ideal sinusoidal wave forms at fundamental frequency, typically close to either 50 or 60 Hz. Harmonics are typically considered a power quality deficiency [1], with increased losses in network and resonance as direct consequences. Indirect consequences include malfunctioning of protection systems and, over time, the additional heating accelerates aging of transformers and lines. Furthermore, motors, invertors and other devices may be sensitive to the presence of distortion.

For these reasons, network operators want to limit harmonics in the network. Electricity supply contracts for large consumers stipulate limits on the amount of harmonic distortion, and penalties when those levels are exceeded. Furthermore, industrial customers may want to limit exposure of sensitive devices to harmonics coming from the network.

Passive and/or active harmonic filters are designed to reduce the presence of harmonics in the network [2]. Passive harmonic filters are composed of series/shunt inductors/capacitors but need to be tuned for specific harmonics, and may change the network resonance. Active harmonic filters are specially-designed power invertors [2], that can optimally compensate harmonics across a range of frequencies without affecting the network resonance. Active harmonic filters have high-bandwidth control loops to inject harmonic currents in anti-phase to offset those present in the network [3]. Putting a harmonic filter at the point of common coupling enables industrial customers to satisfy power quality limits, even when on-site devices produce high levels of distortion. Alternatively, for industrial customers with a significant network behind the meter, putting harmonic filters closer to the sources of the distortion can help with reliability, as it avoids unnecessarily exposing other devices to the harmonics. If there are multiple active filters available across a site, they can be coordinated to provide optimal compensation, using a physics-based optimization model. Prodanovic et al. [4] discuss harmonic compensation by DER as a service, to solve congestion issues in public networks.

B. Literature Review on Harmonic Power Flow Optimization

Two papers by Xia and Heydt focus on the development of simulation codes for harmonic power flow [5], [6]. Dugui and Zheng describe a general harmonic model for single-phase transformers [7]. Dommel et al. propose an approach to include transformer saturation in harmonic power flow simulation through a specialised algorithm [8]. Semlyen et al. use Dommel et al.'s transformer model, and develop an approach to include it in *Newton-type* harmonic power flow solvers [9].

Harmonic impedances for lines and cables can be obtained from first principles, for instance solving Carson's equations at different frequencies [10], or solving Maxwell's equations numerically through a finite elements method [11]. The recent implementation of a harmonic power flow solver in an open-source toolbox in a high-level language is discussed in [12].

Very few studies have explored harmonics in power flow optimization problems at a similar level of detail. Ying-Yi Hong propose a two-level algorithm for harmonic optimal power flow (HOPF) based on generalized Benders decomposition, including MW losses, total harmonic distortion, and current/voltage limits [13]. Wang et al. compare nonlinear current-voltage and power-lifted-voltage semi-definite programming (SDP) models of HOPF [14]. Zhang et al. also develop a SDP relaxation of HOPF [15]. None of these papers discuss the transformer models. Finally, Norouzi et al. project harmonic

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TABLE I
HARMONIC CONTENT OF THE EXCITATION CURRENT FOR MODERATE EXCITATION, RELATIVE TO THE FUNDAMENTAL MAGNETIZING CURRENT.

harmonic number h	3	5	7	9	11
relative magnitude [-]	0.20	0.06	0.02	0.007	0.003

losses onto a fundamental-only optimal power flow (OPF) problem to improve locational marginal pricing [16].

Equivalent OPF formulations can be developed in either power-voltage or rectangular current-voltage (IVR) variables [17]. It should be noted that this remains true in the context of HOPF, because the interaction of currents and voltages at different frequencies does not produce average power [18].

The particular challenge studied in this paper is compensating harmonics across different voltage levels. In contrast to lines and cables which only suffer from distortion, transformers can *cause* harmonic distortion, when operating in saturation. Saturation occurs when an external magnetic field can not further increase the magnetization of the transformer core, resulting in the magnetic flux levelling off. This in turn changes the waveform shape and therefore its spectral content. Table I illustrates the harmonic content of excitation current due to moderate saturation [19].

Transformer saturation is naturally described in the time domain [8], and therefore requires novel treatment to be included in frequency-domain based power flow *optimization* problems. However, the technique developed in [8] is not easily incorporated into an optimization model. If harmonic filters are installed at different voltage levels throughout a power system, the explicit modelling of transformer harmonics is unavoidable for optimal and accurate coordination.

Harmonic loads are produced by non-linear elements such as rectifiers, discharge lighting, or saturated magnetic devices, and can be modeled in a variety of ways [20]. In the realm of fundamental frequency, 'ZIP' and 'exponential' load models, which characterize the voltage-dependence of the power consumption, are commonly used in simulation and have been extended to OPF and its relaxations [21]. Brunoro et al. discuss a variety of load models for higher harmonics [20]:

- a fixed current model, where each harmonic is represented as a constant current source;
- an harmonically coupled admittance matrix model, which relates voltages and currents of several harmonics;
- Norton model, in which each current harmonic is represented by its Norton equivalent; and,
- a hybrid of the previous two.

To keep this paper focused, the constant current model is used consistently for the higher harmonics. For the fundamental, the constant power load model is chosen, to be consistent with canonical OPF.

C. Contributions and Outline of the Paper

The literature review shows that there is limited research on HOPF, and that nonlinear transformer models present a challenge. Therefore, in this work, a methodology is proposed to extend HOPF with models of transformers with excitation. The back-and-forth between time and frequency domain representations will be calculated independently of the optimization procedure through the novel application of a continuous nonlinear fit (splines), similar to the empiric nonlinear model for battery cells in [22]. Concretely, the contributions of the paper include:

1) IVR formulation of HOPF, with clear separation of model and algorithm;
2) three-phase transformer model including the effects of configuration and saturation;
3) two well-described test cases.

The remainder of the paper is organized as follows: Section II presents the mathematical framework, including theoretical background on harmonic (optimal) power flow in the IVR formulation. Section III presents a harmonic model for a three-phase power transformer. Section IV shows numerical results for a prototypical and a real-world example. Section V concludes the paper.

II. MATHEMATICAL FRAMEWORK

This section presents the mathematical framework for HOPF of a phase-balanced three-phase power system. To this end, the mathematical model for power flow is reviewed for both the fundamental and harmonic frequencies. Subsequently, based on the harmonic power flow model, the feasible set of the HOPF is constructed in the IVR variable space.

A. Mathematical Model for Fundamental Frequency

Consider a purely sinusoidal voltage wave at node $i \in \mathcal{I}$ with angular frequency $\omega = 2\pi f$,

$$u_i(t) \stackrel{\text{def}}{=} \sqrt{2} U_i^{\text{mag}} \cos(\omega t + \theta_i), \tag{1}$$

where U_i^{mag} and θ_i respectively denote the *rms* magnitude and angle of the voltage at node i. After phasor transform, the wave is equivalently represented as the rms complex scalar,

$$U_i \stackrel{\text{def}}{=} U_i^{\text{mag}} \angle \theta_i. \tag{2}$$

A tuple $lij \in \mathcal{T}^{1\rightarrow}$ links a branch l to its from-i and to-bus j; $\mathcal{T}^{1\leftarrow}$ contains the equivalent tuples with i and j reversed, and the union set is $\mathcal{T}^1 = \mathcal{T}^{1\rightarrow} \cup \mathcal{T}^{1\leftarrow}$. Branches $l \in \mathcal{L}$ are represented by a π-model (Fig. 1). The series impedance of branch lij is,

$$z_l^{\text{s}} \stackrel{\text{def}}{=} r_l^{\text{s}} + j x_l^{\text{s}}. \tag{3}$$

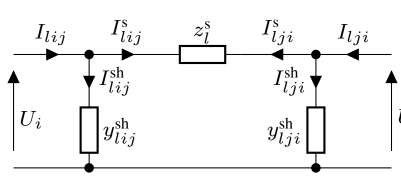


Fig. 1. π-model of branch lij, with series impedance z_l^{s} and from- and to-side shunt admittances y_{lij}^{sh} and y_{lji}^{sh}.

Subsequently, Ohm's law gives,

$$U_i - U_j = z_l^{\mathrm{s}} I_{lij}^{\mathrm{s}}, \tag{4}$$

where I_{lij}^{s} denotes the series current through branch l in the direction $i \rightarrow j$. The **sh**unt admittance of branch lij at its sending and receiving end are,

$$y_{lij}^{\mathrm{sh}} \stackrel{\mathrm{def}}{=} g_{lij}^{\mathrm{sh}} + jb_{lij}^{\mathrm{sh}}, \text{ and } y_{lji}^{\mathrm{sh}} \stackrel{\mathrm{def}}{=} g_{lji}^{\mathrm{sh}} + jb_{lji}^{\mathrm{sh}}. \tag{5}$$

Consequently, the branch shunt currents are,

$$I_{lij}^{\mathrm{sh}} = y_{lij}^{\mathrm{sh}} U_i, \text{ and } I_{lji}^{\mathrm{sh}} = y_{lji}^{\mathrm{sh}} U_j, \tag{6}$$

and applying Kirchhoff's current law (KCL) gives,

$$I_{lij} = I_{lij}^{\mathrm{s}} + I_{lij}^{\mathrm{sh}}, \text{ and } I_{lji} = I_{lji}^{\mathrm{s}} + I_{lji}^{\mathrm{sh}}. \tag{7}$$

Note that this notation implies $I_{lij}^{\mathrm{s}} - I_{lji}^{\mathrm{s}} = 0$. The complex power into branch lij at node i is,

$$S_{lij} \stackrel{\mathrm{def}}{=} U_i I_{lij}^*. \tag{8}$$

Units $u \in \mathcal{U}$ are defined to generalize loads, generators, and storage. Tuple $ui \in \mathcal{T}^{\mathrm{u}}$ link a unit u to node i. The current flowing from node i to unit u is denoted as I_u. Consequently, the complex power consumed by unit u is,

$$S_u \stackrel{\mathrm{def}}{=} U_i I_u^*. \tag{9}$$

Bus shunts are represented as an admittance,

$$y_i^{\mathrm{sh}} \stackrel{\mathrm{def}}{=} g_i^{\mathrm{sh}} + jb_i^{\mathrm{sh}}. \tag{10}$$

Finally, applying KCL at node i gives,

$$\sum_{ui \in \mathcal{T}^{\mathrm{u}}} I_u + \sum_{lij \in \mathcal{T}^{\mathrm{l}}} I_{lij} + y_i^{\mathrm{sh}} U_i = 0. \tag{11}$$

B. Mathematical Model for Harmonic Frequencies

In reality, voltage waves are not necessarily purely sinusoidal, but rather are, more generically, periodic waves. The Fourier transform may be used to decompose periodic waves into the superposition of sinusoidal waves at multiples of the fundamental frequency, i.e., harmonics. A harmonic is a voltage or current at a multiple $h \in \mathcal{H} \subset \mathbb{N}$ of the fundamental frequency. In a balanced three-phase power system, harmonics are classified into three types:

1) positive sequence $h \in \mathcal{H}^{+} \subset \mathcal{H}$: $h = 1 + 3n$,
2) negative sequence $h \in \mathcal{H}^{-} \subset \mathcal{H}$: $h = 2 + 3n$, and
3) zero sequence $h \in \mathcal{H}^{0} \subset \mathcal{H} : h = 3 + 3n$,

where $n \in \mathbb{N}_0$. A property of the Fourier transform is that functions which have half-wave symmetry only contain nonzero magnitudes for the odd harmonics. Therefore, commonly, only odd-order harmonics, i.e., $h \in \{1, 3, 5, ...\}$ are present in a balanced three-phase power systems. Even-order harmonics, i.e., $h \in \{0, 2, 4, 6, ...\}$ are only present given half-wave or defective full-wave rectifiers. In a balanced three-phase power system, the zero-sequence harmonics are in phase with each other and therefore produce a significant zero-sequence neutral current, provided there is a path for that current to flow. These effects are most prominent in transformers depending on their configuration, e.g., non-earthed and delta windings enable blocking of zero-sequence harmonics.

Decoupling of harmonics enables scalable models and algorithms, as it induces sparsity in the system of equations. In order to decouple, i.e., build a mathematical model similar to the one presented in Section II-A but replicated for the harmonic number, two features are exploited: 1) the overall active/reactive power is the sum of the individual harmonic active/reactive powers; and 2) the circuit laws can be decoupled.

1) Power Variables: Consider a general voltage wave $u_i(t)$ at node i at time t, decomposed into harmonics $\mathcal{H}$,

$$u_i(t) \stackrel{\mathrm{def}}{=} U_i^{\mathrm{dc}} + \sum_{h \in \mathcal{H}_0} \sqrt{2} U_{i,h}^{\mathrm{mag}} \cos(h\omega t + \theta_{i,h}), \tag{12}$$

where U_i^{dc}, $U_{i,h}^{\mathrm{mag}}$ and $\theta_{i,h}$ respectively denote the dc-component, i.e., zeroth harmonic, the rms magnitude, and the phase angle of the voltage wave of harmonic $h \in \mathcal{H}$. Using phasor transform, the voltage wave of harmonic $h \in \mathcal{H}_0 = \mathcal{H} \backslash \{0\}$ is represented as a complex rms scalar,

$$U_{i,h} \stackrel{\mathrm{def}}{=} U_{i,h}^{\mathrm{mag}} \angle \theta_{i,h}. \tag{13}$$

Similarly, consider a general current wave $i_e(t)$ into element e at node i, e.g., branch $lij \in \mathcal{T}^{\mathrm{l}}$ or unit $ui \in \mathcal{T}^{\mathrm{u}}$, decomposed in harmonics $\mathcal{H}$:

$$i_e(t) \stackrel{\mathrm{def}}{=} I_e^{\mathrm{dc}} + \sum_{h \in \mathcal{H}_0} \sqrt{2} I_{e,h}^{\mathrm{mag}} \cos(h\omega t + \varphi_{e,h}), \tag{14}$$

where I_e^{dc}, $I_{e,h}^{\mathrm{mag}}$ and $\varphi_{e,h}$ respectively denote the dc-component of the zeroth harmonic, the rms magnitude and the phase angle of the current wave of harmonic. Using phasor transform, the current wave of each harmonic is represented as a complex scalar:

$$I_{e,h} \stackrel{\mathrm{def}}{=} I_{e,h}^{\mathrm{mag}} \angle \varphi_{e,h}. \tag{15}$$

The interaction of currents and voltages at different frequencies does not produce average power [18] (§16.6). The average power consumed by element e at node i over a single period T is derived as:

$$\begin{aligned} P_{i,e} &\stackrel{\mathrm{def}}{=} \frac{1}{T} \int_{t_0}^{t_0+T} u_i(t) i_e(t) \mathrm{d}t \\ &= U_i^{\mathrm{dc}} I_e^{\mathrm{dc}} + \sum_{h \in \mathcal{H}_0} U_{i,h}^{\mathrm{mag}} I_{e,h}^{\mathrm{mag}} \cos(\theta_{i,h} - \varphi_{e,h}) \stackrel{\mathrm{def}}{=} \sum_{h \in \mathcal{H}} P_{i,e,h}. \end{aligned} \tag{16}$$

Equivalently, (16) relates to the complex current (15) and voltage (13) scalars,

$$P_{i,e} = \Re\Big(\sum_{h \in \mathcal{H}} U_{i,h} I_{e,h}^* \Big) = \sum_{h \in \mathcal{H}} P_{i,e,h}. \tag{17}$$

Consequently, the complex harmonic active power into element e at node i is defined for each individual harmonic h,

$$P_{i,e,h} \stackrel{\mathrm{def}}{=} \Re(U_{i,h} I_{e,h}^*), \ Q_{i,e,h} \stackrel{\mathrm{def}}{=} \Im(U_{i,h} I_{e,h}^*). \tag{18}$$

For the purpose of this paper, it is assumed that dc components are negligible, i.e., $\mathcal{H} = \mathcal{H}_0$.

2) Circuit Laws: Linearity is one of the basic properties of the Fourier transform 'F':

$$F\left(af(x)+bg(x)\right)=aF\left(f(x)\right)+bF(g(x))\,, \tag{19}$$

where $f(x)$, $g(x)$ are integrable functions, and $a,b\in\mathbb{C}$. Consequently, Ohm's law and KCL are decoupled:

$$(4)\quad\rightarrow\quad U_{i,h}-U_{j,h}=z^{\text{s}}_{l,h}I^{\text{s}}_{lij,h}, \tag{20}$$
$$(6)\quad\rightarrow\quad I^{\text{sh}}_{lij,h}=y^{\text{sh}}_{lij,h}U_{i,h}, \tag{21}$$
$$(7)\quad\rightarrow\quad I_{lij,h}=I^{\text{s}}_{lij,h}+I^{\text{sh}}_{lij,h}, \tag{22}$$
$$(11)\quad\rightarrow\quad \sum_{ui\in\mathcal{T}^{\text{u}}}I_{u,h}+\sum_{lij\in\mathcal{T}^{\text{l}}}I_{lij,h}+y^{\text{sh}}_{i,h}U_{i,h}=0. \tag{23}$$

The harmonic series impedance of branch lij is,

$$z^{\text{s}}_{l,h}\stackrel{\text{def}}{=}r^{\text{s}}_{l,h}+jx^{\text{s}}_{l,h}. \tag{24}$$

The harmonic shunt admittances of a branch lij are,

$$y^{\text{sh}}_{lij,h}\stackrel{\text{def}}{=}g^{\text{sh}}_{lij,h}+jb^{\text{sh}}_{lij,h},\ \text{and}\ y^{\text{sh}}_{lji,h}\stackrel{\text{def}}{=}g^{\text{sh}}_{lji,h}+jb^{\text{sh}}_{lji,h}. \tag{25}$$

The π-section parameters change significantly and non-linearly depending on harmonic number h. For example, the series resistance depends on the frequency due to the skin effect. A key method for obtaining resistance and reactance across harmonics is solving Carson's equations [10] for different frequencies, given the geometry and wire type of the branch, and then applying the positive-sequence approximation.

Lastly, bus shunts are represented as a harmonic admittance,

$$y^{\text{sh}}_{i,h}\stackrel{\text{def}}{=}g^{\text{sh}}_{i,h}+jb^{\text{sh}}_{i,h}. \tag{26}$$

C. Bounds on Harmonic Distortion

Limits on harmonics are defined in power quality standards such as EN50160 [1], IEC 61000-2-4 [23] and IEC 61000-3-6 [24]. The root mean square (RMS) voltage in the frequency domain is,

$$U^{\text{min}}_i\leq \text{RMS}_i\stackrel{\text{def}}{=}\sqrt{\sum_{h\in\mathcal{H}}|U_{i,h}|^2}\leq U^{\text{max}}_i. \tag{27}$$

The total harmonic distortion (THD) of the voltage is defined as,

$$\text{THD}_i\stackrel{\text{def}}{=}\sqrt{\frac{\sum_{h\in\mathcal{H}\setminus\{1\}}|U_{i,h}|^2}{|U_{i,1}|^2}}\leq \text{THD}^{\text{max}}_i. \tag{28}$$

The individual harmonic distortion (IHD) of the voltage for each specific harmonic h is,

$$\text{IDH}_{i,h}\stackrel{\text{def}}{=}\sqrt{\frac{|U_{i,h}|^2}{|U_{i,1}|^2}}\leq \text{IDH}^{\text{max}}_{i,h}. \tag{29}$$

Equation (27) ensures that the RMS voltage magnitude at node i is bounded between U^{min}_i and U^{max}_i; (28) ensures that the total harmonic distortion remains below a specified limit $\text{THD}^{\text{max}}_i$; and (29) ensures that the amplitude of a given harmonic voltage does not exceed a given percentage $\text{IDH}^{\text{max}}_h$ of the corresponding fundamental, e.g., as stipulated in Tables 1, 4 or 7 of Std. EN50160 [1].

D. IVR Formulation of HOPF

Given the mathematical framework presented in Section II-B, the real-value feasible set for HOPF in the IVR formulation is established:

Reference bus – $\forall i\in\mathcal{I}^{\text{ref}}$:

$$\mathsf{U}^{\text{re}}_{i,h}=0,\ \forall h\in\mathcal{H}\setminus\{1\}, \tag{30}$$
$$\mathsf{U}^{\text{im}}_{i,h}=0,\ \forall h\in\mathcal{H}, \tag{31}$$

KCL (23) – $\forall i\in\mathcal{I},h\in\mathcal{H}$:

$$\sum_{lij\in\mathcal{T}^{\text{l}}}\mathsf{I}^{\text{re}}_{lij,h}+\sum_{ui\in\mathcal{T}^{\text{u}}}\mathsf{I}^{\text{re}}_{ui,h}+g^{\text{sh}}_{i,h}\mathsf{U}^{\text{re}}_{i,h}-b^{\text{sh}}_{i,h}\mathsf{U}^{\text{im}}_{i,h}=0, \tag{32}$$
$$\sum_{lij\in\mathcal{T}^{\text{l}}}\mathsf{I}^{\text{im}}_{lij,h}+\sum_{ui\in\mathcal{T}^{\text{u}}}\mathsf{I}^{\text{im}}_{ui,h}+g^{\text{sh}}_{i,h}\mathsf{U}^{\text{im}}_{i,h}+b^{\text{sh}}_{i,h}\mathsf{U}^{\text{re}}_{i,h}=0, \tag{33}$$

Ohm's law (20) – $\forall lij\in\mathcal{T}^{\text{l}\rightarrow},h\in\mathcal{H}$:

$$\mathsf{U}^{\text{re}}_{j,h}=\mathsf{U}^{\text{re}}_{i,h}-r^{\text{s}}_{l,h}\mathsf{I}^{\text{s,re}}_{lij,h}+x^{\text{s}}_{l,h}\mathsf{I}^{\text{s,im}}_{lij,h}, \tag{34}$$
$$\mathsf{U}^{\text{im}}_{j,h}=\mathsf{U}^{\text{im}}_{i,h}-r^{\text{s}}_{l,h}\mathsf{I}^{\text{s,im}}_{lij,h}-x^{\text{s}}_{l,h}\mathsf{I}^{\text{s,re}}_{lij,h}, \tag{35}$$

Branch total current (22) + (21) – $\forall lij\in\mathcal{T}^{\text{l}},h\in\mathcal{H}$:

$$\mathsf{I}^{\text{re}}_{lij,h}=g^{\text{sh}}_{lij,h}\mathsf{U}^{\text{re}}_{i,h}-b^{\text{sh}}_{lij,h}\mathsf{U}^{\text{im}}_{i,h}+\mathsf{I}^{\text{s,re}}_{lij,h}, \tag{36}$$
$$\mathsf{I}^{\text{im}}_{lij,h}=g^{\text{sh}}_{lij,h}\mathsf{U}^{\text{im}}_{i,h}+b^{\text{sh}}_{lij,h}\mathsf{U}^{\text{re}}_{i,h}+\mathsf{I}^{\text{s,im}}_{lij,h}, \tag{37}$$

The voltage source at the reference bus has idealized characteristics (30), i.e., all voltage harmonics except for the fundamental are zero, and the voltage angle of the fundamental is set to zero. Note that all these constraints are decoupled in h.

Any unit u is either **f**ixed or **d**ispatchable belonging to subsets $\mathcal{U}^{\text{f}}/\mathcal{T}^{\text{u,f}}$ or $\mathcal{U}^{\text{d}}/\mathcal{T}^{\text{u,d}}$, respectively. Dispatchable units, e.g. generators or demand response, are defined with limits $P^{\text{min}}_{u,h}$, $P^{\text{max}}_{u,h}$, $Q^{\text{min}}_{u,h}$ and $Q^{\text{max}}_{u,h}$,

$\forall ui\in\mathcal{T}^{\text{u,d}},h\in\mathcal{H}$:

$$P^{\text{min}}_{u,h}\leq \mathsf{P}_{u,h}\stackrel{\text{def}}{=}\mathsf{U}^{\text{re}}_{i,h}\mathsf{I}^{\text{re}}_{u,h}+\mathsf{U}^{\text{im}}_{i,h}\mathsf{I}^{\text{im}}_{u,h}\leq P^{\text{max}}_{u,h}, \tag{38}$$
$$Q^{\text{min}}_{u}\leq \mathsf{Q}_{u,h}\stackrel{\text{def}}{=}\mathsf{U}^{\text{im}}_{i,h}\mathsf{I}^{\text{re}}_{u,h}-\mathsf{U}^{\text{re}}_{i,h}\mathsf{I}^{\text{im}}_{u,h}\leq Q^{\text{max}}_{u,h}. \tag{39}$$

At the fundamental frequency, the fixed load model is constant power, using set point $P^{\text{ref}}_u+jQ^{\text{ref}}_u$,

$$\mathsf{U}^{\text{re}}_{i,1}\mathsf{I}^{\text{re}}_{u,1}+\mathsf{U}^{\text{im}}_{i,1}\mathsf{I}^{\text{im}}_{u,1}=P^{\text{ref}}_{u,1},\ \forall ui\in\mathcal{T}^{\text{u,f}}, \tag{40}$$
$$\mathsf{U}^{\text{im}}_{i,1}\mathsf{I}^{\text{re}}_{u,1}-\mathsf{U}^{\text{re}}_{i,1}\mathsf{I}^{\text{im}}_{u,1}=Q^{\text{ref}}_{u,1},\ \forall ui\in\mathcal{T}^{\text{u,f}}. \tag{41}$$

For the higher harmonics, the load model is constant current, with a ratio $r_{u,h}$ relative to the current of the fundamental,

$$\mathsf{I}^{\text{re}}_{u,h}+j\mathsf{I}^{\text{im}}_{u,h}=r_{u,h}\left(\mathsf{I}^{\text{re}}_{u,1}+j\mathsf{I}^{\text{im}}_{u,1}\right),\ \forall ui\in\mathcal{T}^{\text{u,f}},h\in\mathcal{H}_0\setminus\{1\}. \tag{42}$$

This constraint links current variables for different h. Variable $\mathsf{W}_{i,h}$ represents the squared voltage magnitude,

$$\mathsf{W}_{i,h}=(\mathsf{U}^{\text{re}}_{i,h})^2+(\mathsf{U}^{\text{im}}_{i,h})^2,\ \forall i\in\mathcal{I},h\in\mathcal{H}. \tag{43}$$

which allows the linear reformulation of the RMS/THD/IDH limits:

$$(27) \rightarrow (U_i^{\min})^2 \leq \sum_{h\in\mathcal{H}_0} \mathsf{W}_{i,h} \leq (U_i^{\max})^2,\ \forall i \in \mathcal{I}, \tag{44}$$

$$(28) \rightarrow \sum_{h\in\mathcal{H}_0\setminus\{1\}} \mathsf{W}_{i,h} \leq (\mathrm{THD}_i^{\mathrm{u}})^2\, \mathsf{W}_{i,1}, \qquad \forall i \in \mathcal{I}, \tag{45}$$

$$(29) \rightarrow \mathsf{W}_{i,h} \leq (\mathrm{IDH}_{i,h}^{\mathrm{u}})^2\, \mathsf{W}_{i,1}, \qquad \forall i \in \mathcal{I}, h \in \mathcal{H}. \tag{46}$$

Finally, the RMS current bound is enforced on branch lij,

$$\sum_{h\in\mathcal{H}_0} \left((\mathsf{I}_{lij,h}^{\mathrm{re}})^2 + (\mathsf{I}_{lij,h}^{\mathrm{im}})^2\right) \leq (I_{lij}^{\mathrm{rated}})^2,\ \forall lij \in \mathcal{T}^1. \tag{47}$$

Note that none of the limit constraints are separable. Therefore, in an optimization context, the solution process cannot be decoupled, due to the linking imposed by those constraints.

III. Harmonic Model of a Power Transformer

In this section, an equivalent single-phase circuit diagram of a two-winding three-phase transformer is presented for the harmonic range considered. This paper builds on the harmonic model for single-phase transformers proposed by Dugui and Zheng [7]. It is extended to two-winding three-phase transformers, including the impact of winding configuration on harmonic power flow and stray capacitance, as discussed in [25], [26], [27]. To facilitate modeling these features, a transformer is split into two parts: the core and windings.

A. Harmonic Model of a Power Transformer Core

Power transformer cores are made of ferromagnetic materials with non-linear magnetization behavior, exhibiting non-linear effects that complicate their analysis, e.g., saturation effects, magnetic hysteresis, and eddy currents. In this work, like Dommel et al. [8], these phenomena are all modeled through the *excitation* current. Transformer excitation current is required to sustain a magnetic field inside the transformer core. The transformer excitation current flows permanently and is largely independent of the secondary load. The transformer excitation current consists of two components: 1) core loss current, and 2) magnetizing current.

1) Core Loss Current: The core loss current represents the resistive core loss and is in phase with the excitation voltage. No-load losses are caused by the flow of excitation current, and includes: (i) iron losses, (ii) copper losses, and (iii) dielectric losses. Only the iron losses caused by eddy currents are significant. No-load losses depend on frequency, maximum flux density and characteristics of magnetic circuit. Consequently, the core loss current is modeled through a equivalent shunt resistance $r_{x,h}^{\mathrm{sh}}$.

2) Magnetizing Current: The waveform of the magnetizing current is constructed based on the waveform of the excitation voltage, through the B-H curve. This time-domain process is illustrated in Fig. 2. Based on the excitation voltage $e(t)$ (top left), the magnetic flux density $B(t)$ (bottom left) is constructed, which is then evaluated for an empirically-derived B-H curve (middle) to obtain the magnetic field intensity $H(t)$

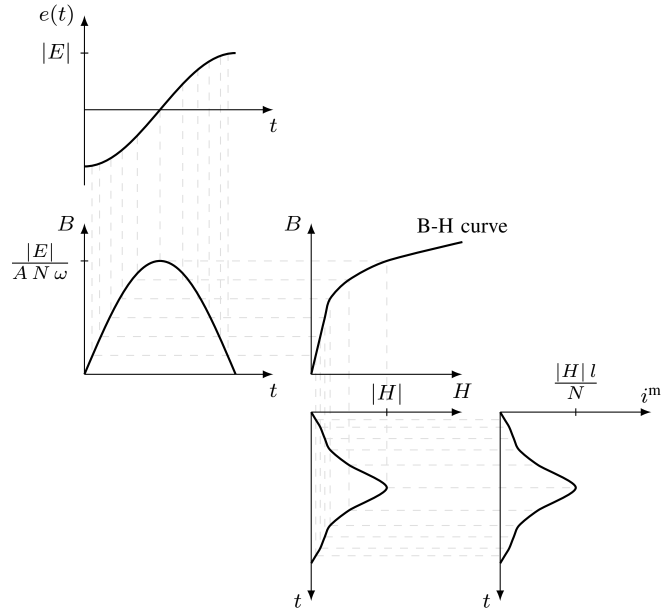


Fig. 2. Illustration of the relationship in the time-domain between the purely-fundamental excitation voltage $e(t)$ and magnetizing current $i^{\mathrm{m}}(t)$, neglecting hysteresis.

(bottom middle), finally resulting in the magnetizing current $i^{\mathrm{m}}(t)$ (bottom right). It is easily seen that the magnetizing current waveform is not sinusoidal and therefore will have harmonic contributions. Even though illustrated in Fig. 2 for a sinusoidal flux, in general, the input waveform can already have harmonic distortion.

In the time domain, the relationship between the magnetizing current $i^{\mathrm{m}}(t)$ and excitation voltage $e(t)$ referred to the primary side is,

$$e(t) = \sum_{h\in H} |E_h| \cdot \sin(\omega_h \cdot t + \theta_h) \tag{48}$$

$$B(t) = \sum_{h\in H} \frac{|E_h|}{A\,N\,\omega_h} \cdot \cos(\omega_h \cdot t + \theta_h) \tag{49}$$

$$H(t) = f^{\mathrm{BH}}\left(B\left(t\right)\right), \tag{50}$$

$$i^{\mathrm{m}}(t) = \frac{l}{N} H(t), \tag{51}$$

where:

- f^{BH} is the B-H curve;
- the effective cross section and length of the transformer core integration path are denoted by A and l; and
- the number of turns of the primary winding is N.

In the context of HOPF, this time domain process needs to be transformed to the frequency domain. Generally, the latter is accomplished through a Fourier transform, however, such a technique is not easily incorporated in an optimization model, and would require specialized algorithmic treatment instead. Therefore, a multi-variate spline approach is proposed instead, similar to the empiric nonlinear model for battery cells in [22]. Even though splines are piece-wise nonlinear

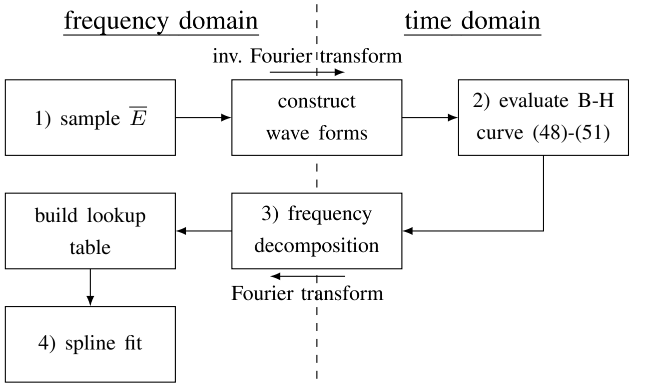


Fig. 3. Process for developing spline fit of magnetizing current harmonics w.r.t. harmonic excitation voltage

functions, inclusion in a nonlinear optimization problem does not necessitate the use of binary variables. Binary variables are required for piece-wise linear functions because those functions are discontinuous at the breakpoints. Conversely, splines are designed to be continuous everywhere within their domain, even at the breakpoints, and consequently pose no obstacle for derivative-based nonlinear optimization algorithms.

The procedure to obtain the necessary splines of the magnetizing current as a function of the excitation voltage is illustrated in Fig. 3 and is composed of the following steps:

1) sample from the space of reasonable excitation voltages E_h in rectangular form for a subset of relevant harmonics $h \in \mathcal{H}^{\text{exc}} \subset \mathcal{H}$;
2) evaluate the B-H curve using the rectangular voltage samples translated to the time domain as $e(t)$ to obtain the corresponding magnetizing current $i^{\text{m}}(t)$ following [8];
3) the resulting magnetizing current $i^{\text{m}}(t)$ is translated to magnetizing current I^{m}_h in rectangular form for all harmonics $h \in \mathcal{H}$;
4) establish a multivariate spline using the resulting look-up table, and;
5) finally, these splines are incorporated into the optimization program as nonlinear constraints,

$$\mathsf{I}^{\text{m,re}}_{x,h} = f_{x,h}(\overline{\mathsf{E}}^{\text{re}}_{x}, \overline{\mathsf{E}}^{\text{im}}_{x}),\ \mathsf{I}^{\text{m,im}}_{h} = f_{x,h}(\overline{\mathsf{E}}^{\text{re}}_{x}, \overline{\mathsf{E}}^{\text{im}}_{x}), \forall h \in \mathcal{H}, \quad (52)$$

where $\overline{\mathsf{E}}^{\text{re}}_{x}$ and $\overline{\mathsf{E}}^{\text{im}}_{x}$ denote the vector of real and imaginary excitation voltages of the relevant harmonics $\mathcal{H}^{\text{exc}}$ of transformer $x \in \mathcal{X}$. Note that this fitting has to happen only once for each transformer x, which can be done prior to building the optimization model, and that the coefficients of the splines can be stored for re-use. Furthermore, there are few restrictions on the description of the B-H curve, which is typically obtained empirically. Both the nonlinearity and the hysteresis can be dealt with, the latter is outside the scope of this paper. The B-H curve can even be discontinuous, as the spline fit makes the end-to-end model continuous.

B. Representation of Transformer Core

In the transformer core Kirchhoff's current law is enforced, accounting for the vector group t^{vg} (Fig. 4). In rectangular

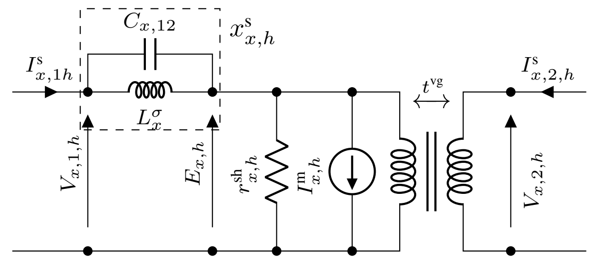


Fig. 4. Equivalent single-phase circuit diagram of the core of a two-winding three-phase transformer $x \in \mathcal{X}$ for a given harmonic $h \in \mathcal{H}$.

coordinates, this is,

$$\mathsf{I}^{\text{m,re}}_{x,h} + \frac{\mathsf{E}^{\text{re}}_{x,h}}{r^{\text{sh}}_{x,h}} = \sum_{w \in \mathcal{W}} t^{\text{vg,re}}_{x,h}\, \mathsf{I}^{\text{s,re}}_{x,w,h} + t^{\text{vg,im}}_{x,h}\, \mathsf{I}^{\text{s,im}}_{x,w,h},\ \forall h \in \mathcal{H}, \quad (53)$$

$$\mathsf{I}^{\text{m,im}}_{x,h} + \frac{\mathsf{E}^{\text{im}}_{x,h}}{r^{\text{sh}}_{x,h}} = \sum_{w \in \mathcal{W}} t^{\text{vg,re}}_{x,h}\, \mathsf{I}^{\text{s,im}}_{x,w,h} - t^{\text{vg,im}}_{x,h}\, \mathsf{I}^{\text{s,re}}_{x,w,h},\ \forall h \in \mathcal{H}, \quad (54)$$

where $t^{\text{vg,re}}_{h}$ and $t^{\text{vg,im}}_{h}$ represents the transformation for a given vector group relative to the primary winding.

Additional voltage variables V and E are included to represent the internal node voltages and excitation voltage, respectively (Fig. 4). The voltage drop over the leakage inductance and stray inter-winding capacitance are,

$$\mathsf{V}^{\text{re}}_{x,h,1} = \mathsf{E}^{\text{re}}_{x,h} - x^{\text{s}}_{x,h}\, \mathsf{I}^{\text{s,im}}_{x,h,1},\ \forall h \in \mathcal{H}, \quad (55)$$

$$\mathsf{V}^{\text{im}}_{x,h,1} = \mathsf{E}^{\text{im}}_{x,h} + x^{\text{s}}_{x,h}\, \mathsf{I}^{\text{s,re}}_{x,h,1},\ \forall h \in \mathcal{H}, \quad (56)$$

where $x^{\text{s}}_{x,h}$ denotes the equivalent reactance of the leakage inductance and stray inter-winding capacitance for harmonic h.

Lastly, the effects of the vector group t^{vg} is reflected at the terminal of the secondary winding,

$$\mathsf{V}^{\text{re}}_{x,h,2} = t^{\text{vg,re}}_{x,h}\, \mathsf{E}^{\text{re}}_{x,h} - t^{\text{vg,im}}_{x,h}\, \mathsf{E}^{\text{im}}_{x,h},\ \forall h \in \mathcal{H}, \quad (57)$$

$$\mathsf{V}^{\text{im}}_{x,h,2} = t^{\text{vg,re}}_{x,h}\, \mathsf{E}^{\text{im}}_{x,h} + t^{\text{vg,im}}_{x,h}\, \mathsf{E}^{\text{re}}_{x,h},\ \forall h \in \mathcal{H}. \quad (58)$$

C. Representation of the Transformer Winding

The possibility of isolating zero-sequence currents for a certain winding configuration, e.g., wye, delta, etc., is captured through the model of the corresponding power transformer winding. Consequently, a distinct equivalent circuit diagram of a winding is needed for each of the positive/negative sequence and zero sequence harmonics, respectively depicted in Fig. 5 and 6. The latter changes depending on the configuration: (a) earthed, (b) non-earthed, or (c) delta winding.

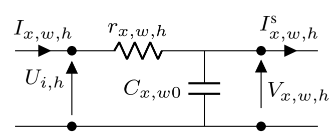
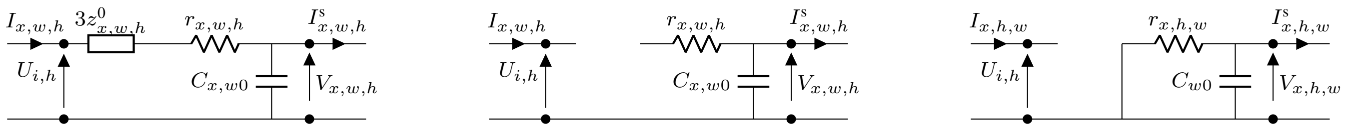


Fig. 5. Equivalent single-phase circuit diagram of a winding $w \in \mathcal{W}$ of a two-winding three-phase transformer for a given positive or negative sequence harmonic $h \in \mathcal{H}^{+} \cup \mathcal{H}^{-}$.



(a) earthed winding $w \in \mathcal{W}^{\mathrm{e}}$ (b) non-earthed winding $w \in \mathcal{W}^{\mathrm{ne}}$ (c) delta winding $w \in \mathcal{W}^{\mathrm{d}}$

Fig. 6. Equivalent single-phase circuit diagram of a winding $w \in \mathcal{W}$ of a two-winding three-phase transformer for a given zero sequence harmonic $h \in \mathcal{H}^0$.

Independent on the winding configuration, Kirchhoff's current law is enforced between the winding and the core,

$$\forall h \in \mathcal{H}, w \in \mathcal{W}:$$
$$\mathsf{I}^{\mathrm{s,re}}_{x,w,h} = \mathsf{I}^{\mathrm{re}}_{x,w,h} + g^{\mathrm{sh}}_{x,w,h}\,\mathsf{V}^{\mathrm{re}}_{x,w,h} - b^{\mathrm{sh}}_{x,w,h}\,\mathsf{V}^{\mathrm{im}}_{x,w,h}, \quad (59)$$
$$\mathsf{I}^{\mathrm{s,im}}_{x,w,h} = \mathsf{I}^{\mathrm{im}}_{x,w,h} + g^{\mathrm{sh}}_{x,w,h}\,\mathsf{V}^{\mathrm{im}}_{x,w,h} + b^{\mathrm{sh}}_{x,w,h}\,\mathsf{V}^{\mathrm{re}}_{x,w,h}, \quad (60)$$

where $g^{\mathrm{sh}}_{x,w,h}$ and $b^{\mathrm{sh}}_{x,w,h}$ represent the stray capacitance $C_{x,w0}$ between winding w and transformer tank. Alternatively, in case of a delta winding for zero-sequence harmonic $h \in \mathcal{H}^0$, $g^{\mathrm{sh}}_{x,w,h}$ and $b^{\mathrm{sh}}_{x,w,h}$ represent the parallel circuit of winding resistance $r_{x,w,h}$ and stray capacitance $C_{x,w0}$ (Fig. 6c).

For positive and negative sequence harmonics $h \in \mathcal{H}^+ \cup \mathcal{H}^-$, Ohm's law over winding $w \in \mathcal{W}$ gives (Fig. 5),

$$\forall h \in \mathcal{H}^+ \cup \mathcal{H}^-, w \in \mathcal{W}: w \rightarrow i \in \mathcal{I}:$$
$$\mathsf{U}^{\mathrm{re}}_{i,h} = \mathsf{V}^{\mathrm{re}}_{x,w,h} + r_{x,w,h}\,\mathsf{I}^{\mathrm{re}}_{x,w,h}, \quad (61)$$
$$\mathsf{U}^{\mathrm{im}}_{i,h} = \mathsf{V}^{\mathrm{im}}_{x,w,h} + r_{x,w,h}\,\mathsf{I}^{\mathrm{im}}_{x,w,h}, \quad (62)$$

where winding resistance $r_{x,w,h}$ is responsible for the Joule losses for a given harmonic h. For zero-sequence harmonics $h \in \mathcal{H}^0$, Ohm's law over earthed winding $w \in \mathcal{W}^{\mathrm{e}} \subset \mathcal{W}$ gives (Fig. 6a),

$$\forall h \in \mathcal{H}^0, w \in \mathcal{W}^{\mathrm{e}}: w \rightarrow i \in \mathcal{I}:$$
$$\mathsf{U}^{\mathrm{re}}_{i,h} = \mathsf{V}^{\mathrm{re}}_{x,w,h} + (r_{x,w,h} + 3r^0_{x,w,h})\,\mathsf{I}^{\mathrm{re}}_{x,w,h} - 3x^0_{x,w,h}\,\mathsf{I}^{\mathrm{im}}_{x,w,h}, \quad (63)$$
$$\mathsf{U}^{\mathrm{im}}_{i,h} = \mathsf{V}^{\mathrm{im}}_{x,w,h} + (r_{x,w,h} + 3r^0_{x,w,h})\,\mathsf{I}^{\mathrm{im}}_{x,w,h} + 3x^0_{x,w,h}\,\mathsf{I}^{\mathrm{re}}_{x,w,h}, \quad (64)$$

where $r^0_{x,w,h}$ and $x^0_{x,w,h}$ denote the neutral resistance and reactance. For zero-sequence harmonics $h \in \mathcal{H}^0$, both a non-earthed and delta winding $w \in (\mathcal{W}^{\mathrm{ne}} \cup \mathcal{W}^{\mathrm{d}}) \subset \mathcal{W}$ prevent zero-sequence currents into the corresponding node i (Fig. 6b and 6c),

$$\forall h \in \mathcal{H}^0, w \in \mathcal{W}^{\mathrm{ne}} \cup \mathcal{W}^{\mathrm{d}}: w \rightarrow i \in \mathcal{I}:$$
$$\mathsf{I}^{\mathrm{re}}_{x,w,h} = 0, \quad (65)$$
$$\mathsf{I}^{\mathrm{im}}_{x,w,h} = 0. \quad (66)$$

The transformer feasible set is (52)-(66). The linking variables w.r.t to the feasible set in §II-D are the bus voltages $\mathsf{U}^{\mathrm{re}}_{i,h}, \mathsf{U}^{\mathrm{im}}_{i,h}$ and the currents $\mathsf{I}^{\mathrm{re}}_{x,w,h}, \mathsf{I}^{\mathrm{im}}_{x,w,h}$ are added to KCL (32)-(33).

IV. Numerical Illustrations

The feasible set is implemented in the algebraic modeling language JuMP [28], and specifically use the user-defined functions feature to evaluate and differentiate the splines and to interface with nonlinear programming solver Ipopt [29]. Calculations ran on an AMD Ryzen 3600 with 16 GB RAM.

A. Harmonic Power Flow Example

A two-bus example network with data for 1st and 3rd harmonics is shown in Fig. 7. The network is composed of an impedance between buses i and j, with a load d on bus j and a generator g on bus i. The ideal slack generator is distortion-free, i.e., only generates nonzero voltage at the fundamental.

Table II present the simulated results for the bus, load, generator and line variables, respectively. As one can see, conservation of current is satisfied, as $I_{lij,h} = I_{g,h} = I_{d,h}$ for $h \in \{1, 3\}$. The load is indeed operating at constant power at the fundamental, as the power set point $\mathsf{P}_{d,h} + j\mathsf{Q}_{d,h}$ matches the target value, despite a voltage drop of 3.3 %. The 3rd harmonic is constant current relative to the fundamental as, $|I_{d,3}|/|I_{d,1}| = 0.2$ equals the given ratio $r_{u,3}$. At bus j, a voltage drop in the fundamental is observed, as well as harmonic distortion, with the 3rd harmonic voltage at 0.023 pu. As the generator on the reference bus is idealized, no distortion occurs on bus i – it can absorb harmonic current without suffering harmonic voltage change. Generation losses are $0.086/4.086 = 2.1\,\%$. The branch losses are 0.086 pu

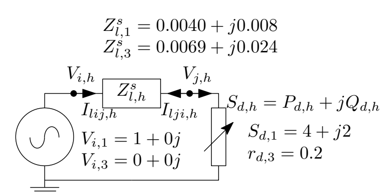


Fig. 7. Two-bus example network in per-unit quantities

TABLE II
Bus results for two-bus line case (per unit).

bus	harmonic	$\lvert U_{i,h}\rvert$	$\angle U_{i,h}$	RMS$_i$	THD$_i$
i	$h=1$	1.000	0	1.000	1.000
	$h=3$	0	0	-	-
j	$h=1$	0.967	-1.423	0.967	0.024
	$h=3$	0.023	-134.090	-	-
load	**harmonic**	$\mathsf{P}_{d,h}$	$\mathsf{Q}_{d,h}$	$I_{d,h}$	$\lvert I_{d,h}\rvert$
d	$h=1$	4.000	2.000	$4.086 - 2.171j$	4.627
	$h=3$	-0.006	-0.021	$0.817 - 0.434j$	0.925
gen.	**harmonic**	$\mathsf{P}_{g,h}$	$\mathsf{Q}_{g,h}$	$I_{g,h}$	$\lvert I_{g,h}\rvert$
g	$h=1$	4.086	2.171	$4.086 - 2.171j$	4.627
	$h=3$	0	0	$0.817 - 0.434j$	0.925
line	**harmonic**	$\mathsf{P}_{lij,h}$	$\mathsf{Q}_{lij,h}$	$I_{lij,h}$	$\mathsf{P}^{\mathrm{loss}}_{l,h}$
l	$h=1$	4.086	2.171	$4.086 - 2.171j$	0.086
	$h=3$	0	0.006	$0.817 - 0.434j$	0.006

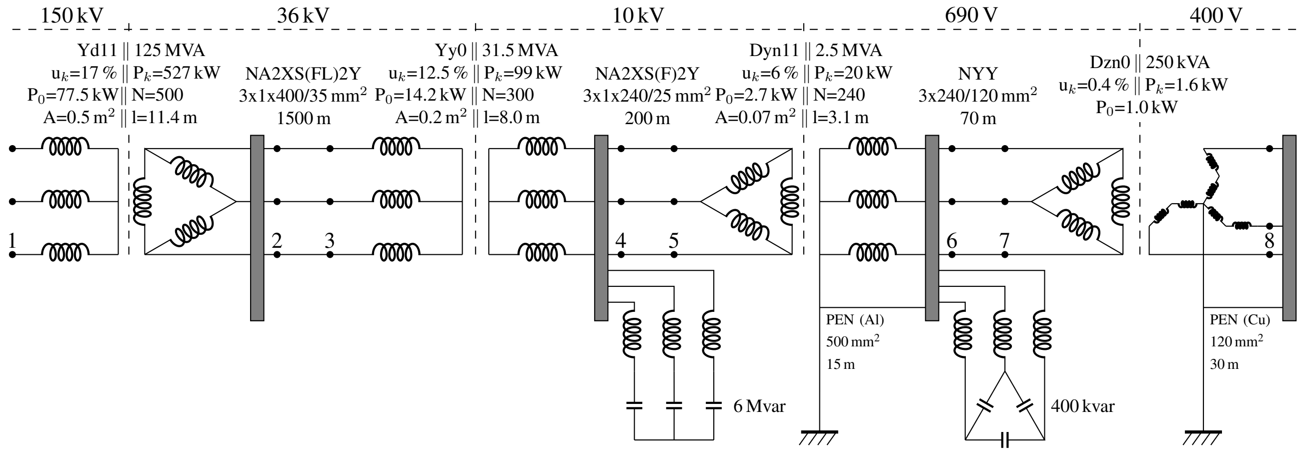


Fig. 8. Real-world network across five voltage levels, with four transformers, three cables, and two capacitor banks.

on the fundamental, and $0.006\,\mathrm{pu}$ due to the third harmonic current, i.e., an increase of the network losses of $7\,\%$ relative to the fundamental.

B. Real-World HOPF Use-Case

This case study considers a real-world network with five voltage levels, four transformers, two capacitor banks and three cables (Fig. 8). The loads are listed in Table III, two of which produce significant harmonics: a six-pulse converter on bus six and fluorescent lighting on bus eight, both with a low power factor. A proof-of-concept study is developed where the installation of a three-phase active filter is considered, rated at 600 A, at bus six (690 V). Table IV lists the external system generator and filter data. The external system generator is an ideal distortion-free slack generator, i.e., only generates nonzero voltage at the fundamental. The active filter is a generator with additional restrictions,

$$\mathsf{P}_{u,1} = 0, \tag{67}$$

$$\sum_{h\in\mathcal{H}} \mathsf{P}_{u,h} = 0. \tag{68}$$

This means that the filter: 1) can control reactive power independently for each harmonic; 2) is lossless (fundamental equals zero); and 3) absorbs/injects higher harmonics as long as the active power cancels out.

On top of the fundamental, the following harmonics $h \in \mathcal{H} = [3, 5, 7, 9, 13]$ are considered with accompanying individual harmonic distortion limits $\mathrm{IDH}_h \in [5.0, 6.0, 5.0, 1.5, 3.0]\,\%$. The voltage THD at bus one is limited to $1.2\,\%$, whereas the other buses are limited to $8.0\,\%$. The RMS voltage in the low-voltage parts is in the range of 0.9 to $1.1\,\mathrm{pu}$, elsewhere it is 0.94 to $1.06\,\mathrm{pu}$.

All transformers are build using core material with a B-H curve as depicted in Fig. 9. Splines are built for the magnetizing current of the first three transformers for all considered harmonics $h \in \mathcal{H}$. To this end, the real and imaginary excitation voltages are sampled for harmonics $h \in \mathcal{H}^{\mathrm{exc}} = \{1, 5\}$ for a range $[-1.1, 1.1]$ and $[-0.066, 0.066]\,\mathrm{pu}$, respectively, with 21 and 11 uniform steps, respectively. The 3rd harmonic is skipped as it is anyway suppressed by the transformer configuration. This is a total of four input dimensions, or $21^2 \cdot 11^2 \approx 53361$ samples. It takes about six minutes per transformer to evaluate the process describe in Fig. 3. For the purpose of this case study, transformer stray capacitance is neglected, all other inputs are derived from the data in Fig. 8.

TABLE III
HARMONIC LOADS IN REAL-WORLD TEST CASE

			rel. current magnitude [%]				
bus	P [MW]	Q [Mvar]	$r_{u,3}$	$r_{u,5}$	$r_{u,7}$	$r_{u,9}$	$r_{u,13}$
2	50.000	15.880	0	0	0	0	0
4	12.600	4.001	0	0	0	0	0
6	1.420	0	0	45	25	0	8
8	0.050	0.040	20	10	5	6	0
sum	64.070	19.920	-	-	-	-	-

TABLE IV
EXTERNAL SYSTEM GENERATOR AND FILTER DATA

unit u	bus	$P_{u,1}^{\max}$ [MW]	$Q_{u,1}^{\max}$ [Mvar]	I_{lij}^{rated} [A]
external system (gen.)	1	∞	∞	∞
filter	6	0.717	0.717	600

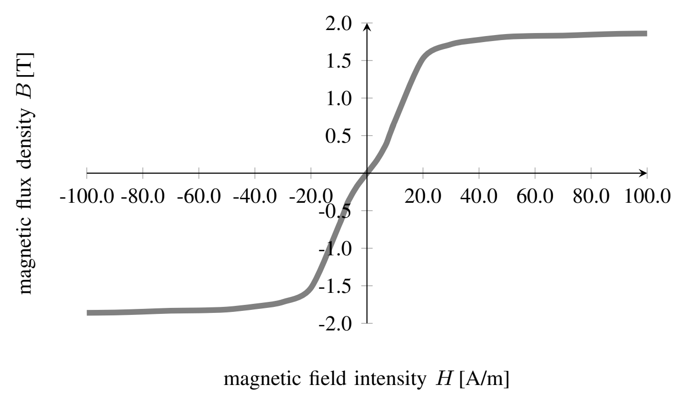


Fig. 9. B-H curve corresponding to the core material used in all transformers.

Two cases are considered: 1) base case harmonic power flow without active filter, and 2) harmonic optimal power flow with active filter. The cases presented are solved in less than one second by Ipopt, including the call-back time for the derivatives. Comparatively, the spline fits take about six minutes to set up for each transformer. Results are presented in a $100\,\text{MVA}$ base.

1) Harmonic Power Flow Without Active Filter: The resulting voltages from the base case power flow are listed in Table V. As expected, given the ideal slack generator on bus one, all corresponding voltages except for the fundamental are zero. Furthermore, the zero-sequence harmonics are indeed absorbed in the delta winding of associated transformers, so their magnitudes are negligible and therefore unlisted. Lastly, as expected, a phase shift of $30°$ occurs over the first and fourth transformer, as both have a phase displacement number of eleven. For the positive sequence harmonics, i.e., $h \in \{1, 7, 13\}$, this result in a primary lead of approximately $30°$, whereas for the negative sequence harmonics, i.e., $h = 5$, a primary lag of approximately $30°$ can be noticed. Given the ideal slack generator on bus one, this effect is only noticeable at the fourth transformer for non-fundamental harmonics.

Table VI highlights some relevant results for the excitation voltages and magnetizing currents of transformers $x \in \{1, 2, 3\}$. Two aspects are highlighted, first, in line with Fig. 2, the fundamental excitation voltage phase is lagging approximately $90°$ to the fundamental magnetizing current phase. Second, the magnitudes of different harmonic magnetizing currents follows the relative magnitudes in Table I, except for the third harmonic which is expected to be larger. This may be attributed to the excitation voltage being lower than one.

Lastly, the harmonic currents imposed on the ideal slack generator at bus one are,

$$\mathsf{I}_{g_1,1} = 0.6435 - j\,0.1682,\ \mathsf{I}_{g_1,5} = -0.0045 - j\,0.006,$$
$$\mathsf{I}_{g_1,7} = 0.0049 - j\,0.0005,\ \mathsf{I}_{g_1,13} = 0.0048 - j\,0.0006,$$

and the corresponding current THD is $1.16\,\%$.

TABLE V
CASE 1 - BUS VOLTAGES [PU/°], 3RD AND 9TH HARMONIC ARE ZERO.

i	$\lvert\mathsf{U}_{i,1}\rvert$	$\angle\mathsf{U}_{i,1}$	$\lvert\mathsf{U}_{i,5}\rvert$	$\angle\mathsf{U}_{i,5}$	$\lvert\mathsf{U}_{i,7}\rvert$	$\angle\mathsf{U}_{i,7}$	$\lvert\mathsf{U}_{i,13}\rvert$	$\angle\mathsf{U}_{i,13}$	THD$_i$
1	1.000	0	0	0	0	0	0	0	0.000
2	0.979	-34.86	0.005	173.58	0.004	-126.35	0.008	-127.27	0.011
3	0.979	-34.87	0.005	173.48	0.004	-126.44	0.008	-127.32	0.011
4	0.979	-35.06	0.006	173.52	0.005	-126.39	0.009	-127.26	0.012
5	0.979	-35.06	0.006	173.51	0.005	-126.41	0.009	-127.26	0.012
6	0.979	-65.07	0.006	-156.63	0.005	-156.50	0.009	-157.27	0.012
7	0.979	-65.07	0.006	-156.62	0.005	-156.50	0.009	-157.27	0.012
8	0.979	–65.07	0.006	-156.62	0.005	-156.50	0.009	-157.27	0.012

TABLE VI
EXCITATION VOLTAGES AND MAGNETIZING CURRENTS [PU/°].

x	$\lvert\mathsf{E}_{x,1}\rvert$	$\angle\mathsf{E}_{x,1}$	$\lvert\mathsf{E}_{x,5}\rvert$	$\lvert\mathsf{I}^{\text{m}}_{x,1}\rvert$	$\angle\mathsf{I}^{\text{m}}_{x,1}$	$\lvert\mathsf{I}^{\text{m}}_{x,3}\rvert$	$\lvert\mathsf{I}^{\text{m}}_{x,5}\rvert$	$\lvert\mathsf{I}^{\text{m}}_{x,7}\rvert$	$\lvert\mathsf{I}^{\text{m}}_{x,9}\rvert$	$\lvert\mathsf{I}^{\text{m}}_{x,13}\rvert$
1	0.980	-4.9	0.005	$4.7e^{-5}$	85.1	$1.4e^{-6}$	$2.6e^{-6}$	$7.1e^{-7}$	$2.8e^{-7}$	$1.8e^{-7}$
2	0.980	-35.1	0.006	$1.3e^{-5}$	55.0	$3.0e^{-7}$	$8.0e^{-7}$	$2.1e^{-7}$	$5.9e^{-8}$	$5.6e^{-8}$
3	0.980	-35.1	0.006	$1.8e^{-5}$	55.0	$4.3e^{-8}$	$1.1e^{-7}$	$2.5e^{-8}$	$9.2e^{-9}$	$6.2e^{-9}$

2) Harmonic Optimal Power Flow with Active Filter: An objective is introduced to minimize the voltage THD at bus six,

$$\min \sum_{h \in \mathcal{H} \setminus \{1\}} \mathsf{W}_{6,h}. \tag{69}$$

The active and reactive power consumed by the active filter is listed in the top half of Table VII. In line with restriction (67), the fundamental active power generated by the active filter is zero. Furthermore, in line with restriction (68), the aggregated active power over all non-fundamental harmonics equals zero. The ideal slack generator is dispatched to $0.64354 + j\,0.16839\,\text{pu}$, with no harmonic contribution.

The resulting voltages for the optimal power flow with active filter are listed in Table VIII. The phase angles for those voltages with a zero magnitude are omitted as they become meaningless. The active filter succeeds in filtering all harmonics currents injected by the six-pulse converter on bus six and fluorescent lighting on bus eight. Consequently, all non-fundamental voltages and the voltage THD approximate zero.

If the HOPF is solved without magnetizing current, i.e., $\mathsf{I}^{\text{m,re}}_{x,h} = \mathsf{I}^{\text{m,im}}_{x,h} = 0$, the ideal slack generator is dispatched at $0.64384 + j0.16849\,\text{pu}$, still without harmonics. The dispatch of the active filter changes considerably (bottom half of Table VII). Again, both restrictions (67) and (68) imposed on the active filter are respected.

V. Conclusions

In this work, a detailed formulation of HOPF in current-voltage variables, extended with transformers, is proposed and tested. Two test cases are illustrated, including a novel real-world test case for which the full network model, load/generation data, and applicable harmonic limits are provided. It is noted that setting up such test cases requires a non-negligible amount of data collection.

TABLE VII
ACTIVE AND REACTIVE POWER CONSUMED BY THE ACTIVE FILTER [PU], WITH AND WITHOUT CONSIDERING MAGNETIZING CURRENT.

		h=1	h=3	h=5	h=7	h=9	h=13
with	$\mathsf{P}_{2,h}$	0.0	$1.70e^{-16}$	$2.95e^{-11}$	$6.31e^{-12}$	$5.65e^{-16}$	$-3.58e^{-11}$
	$\mathsf{Q}_{2,h}$	$2.55e^{-5}$	$6.34e^{-15}$	$1.73e^{-10}$	$1.04e^{-10}$	$4.07e^{-14}$	$-3.15e^{-11}$
w/o	$\mathsf{P}_{2,h}$	0.0	$1.14e^{-10}$	$8.17e^{-8}$	$-1.90e^{-7}$	$2.26e^{-11}$	$1.08e^{-7}$
	$\mathsf{Q}_{2,h}$	$-2.54e^{-4}$	$1.52e^{-7}$	$6.58e^{-7}$	$-4.19e^{-7}$	$5.06e^{-8}$	$6.92e^{-9}$

TABLE VIII
CASE 2 - BUS VOLTAGES [PU/°], 3RD AND 9TH HARMONIC ARE ZERO.

i	$\lvert\mathsf{U}_{i,1}\rvert$	$\angle\mathsf{U}_{i,1}$	$\lvert\mathsf{U}_{i,5}\rvert$	$\angle\mathsf{U}_{i,5}$	$\lvert\mathsf{U}_{i,7}\rvert$	$\angle\mathsf{U}_{i,7}$	$\lvert\mathsf{U}_{i,13}\rvert$	$\angle\mathsf{U}_{i,13}$	THD$_i$
1	1.000	0	0	0	0	0	0	0	0
2	0.979	-34.86	0.000	-	0.000	-	0.000	-	$4.2e^{-7}$
3	0.979	-34.87	0.000	-	0.000	-	0.000	-	$4.2e^{-7}$
4	0.979	-35.06	0.000	-	0.000	-	0.000	-	$1.4e^{-7}$
5	0.979	-35.06	0.000	-	0.000	-	0.000	-	$1.4e^{-7}$
6	0.979	-65.07	0.000	-	0.000	-	0.000	-	$9.0e^{-7}$
7	0.979	-65.07	0.000	-	0.000	-	0.000	-	$8.9e^{-7}$
8	0.979	–65.07	0.000	-	0.000	-	0.000	-	$1.6e^{-7}$

The transformer's nonlinear magnetizing current is represented through a multivariate spline fit, directly linking it to the transformer excitation voltage directly, *without* passing through the time domain during the optimization phase. The scalability in terms of the number of excitation voltage harmonics targeted is nevertheless inherently limited due to the curse of dimensionality, because of the sampling requirement in many dimensions. The spline fit can still be improved, e.g., by exploiting better sampling in different coordinates (polar vs rectangular) or by performing non-uniform sampling focusing on regions of interest. The proposed transformer models also work without B-H curves, with the magnetizing current set to zero. The differences between both approaches are small in the proposed test cases, but further exploration on a broader set of test cases is needed to make this observation general.

The availability of test cases for HOPF is very limited, and PGLIB [30] could be a community to spearhead the development of such test cases. HOPF with explicit *phase unbalance* could be particularly interesting to analyse opportunities offered by single-phase ac electric vehicle charging vs three-phase dc fast charging, where the off-board chargers could even function as active filters.

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[1]available at github.com/JuliaDynamics/SignalDecomposition.jl
[2]available at github.com/JuliaMath/Interpolations.jl